Collatz prefixes with finitely many deviations from rational periodic trajectories

Ethan Walker*

21 October 2026

Abstract

For the accelerated odd Collatz map, a prefix with at most R divisions by 4 or more has length at most $(1+\varepsilon)\log_2(n+1)+O_{R,\varepsilon}(1)$, for each fixed R and $\varepsilon > 0$. We prove a common generalization: the same leading coefficient bounds reference depth for prefixes with finitely many deviations from any fixed finite collection of rational periodic trajectories. The proof uses the p-adic Subspace Theorem and finiteness of nondegenerate S-unit equations. The constants are ineffective and not uniform in the number of deviations.

1 The simplest statement

For a positive odd integer n, put

$$a(n) = v_2(3n+1), \qquad U(n) = \frac{3n+1}{2^{a(n)}}.$$

A step is *deep* if $a(n) \geq 2$; otherwise it is shallow.

Corollary 1. *For every fixed integer $R \geq 0$ and every $\varepsilon > 0$, there is a finite constant $K_{R,\varepsilon}$ such that a prefix $n_0, \ldots, n_h$ with at most R deep steps satisfies*

$$h \leq (1+\varepsilon)\log_2(n_0+1) + K_{R,\varepsilon}.$$

The leading coefficient 1 cannot be decreased: starting at $n_0 = 2^{h+1}-1$ gives h shallow steps. For $R = 0$ the bound follows directly from $n_t + 1 = (3/2)^t(n_0+1)$ and divisibility by 2^{h+1}. The work below allows any fixed number of deep steps, whose depths can be arbitrarily large.

2 A finite periodic dictionary

Extend U to odd rational 2-adic integers using the same valuation rule. Fix a finite nonempty union C of complete periodic trajectories of this map. Write

$$b(c) = v_2(3c+1) \geq 1, \qquad g(c) = \frac{3c+1}{2^{b(c)}} \in C.$$

These rationals have denominators coprime to 6: the period equation has denominator dividing $2^A - 3^m$. All valuations below are well defined, and c and $3c+1$ are nonzero. A reference path is any $c_0, \ldots, c_h \in C$. Its step t needs a *repair* if

$$a(n_t) \neq b(c_t) \quad \text{or} \quad c_{t+1} \neq g(c_t).$$

*Prepared from the author's October 2026 research notes with OpenAI ChatGPT/Codex assistance, including proof reconstruction and exposition. This preprint has not undergone independent human peer review.

A change of reference phase also costs a repair. Define the reference depth $B_h = \sum_{t=0}^{h-1} b(c_t)$, and let β be the smallest mean of b over the cycles in C.

Theorem 2. *Fix C, $R \geq 0$, and $\varepsilon > 0$. Suppose a positive odd orbit prefix has a reference path with at most R repairs. Assume either all members of C are negative, or*

$$n_t \geq L_C := \left\lceil 2 \max_{c \in C} |c| \right\rceil + 2 \quad (0 \leq t \leq h).$$

Then constants depending only on C, R, ε satisfy

$$B_h \leq (1+\varepsilon) \log_2(n_0 + 1) + K, \qquad h \leq \frac{1+\varepsilon}{\beta} \log_2(n_0 + 1) + K'.$$

Corollary 1 is $C = \{-1\}$: here $b = 1$ and $g(-1) = -1$. The size condition matters for positive references: the actual fixed point 1 could otherwise follow itself forever.

3 Two standard Diophantine inputs

Subspace Theorem. Fix a finite set of places S of $\mathbb{Q}$ containing infinity. At each place choose d independent rational linear forms in d variables. For every $\eta > 0$, the integer vectors $X \neq 0$ satisfying

$$\prod_{v \in S} \prod_{j=1}^{d} |L_{j,v}(X)|_v \leq \|X\|_\infty^{-\eta}$$

lie in finitely many proper rational linear subspaces. This is the rational case of the p-adic theorem in Bilu [1, Theorem 2.3].

S-unit finiteness. A minimal zero sum of a fixed number of terms, each a fixed nonzero rational coefficient times an element of a fixed finitely generated subgroup of $\mathbb{Q}^*$, has finitely many projective term ratios. Divide by one term and apply Evertse–Schlickewei–Schmidt [2, Theorem 1.1]. We use the group generated by 2 and 3.

4 Proof of Theorem 2

Choose an odd common denominator D for C, coprime to 3. Put

$$a_t = a(n_t), \quad A_t = \sum_{j<t} a_j, \quad Z_t = \frac{2^{A_t}}{3^t}(n_t - c_t) > 0.$$

A good step leaves Z_t unchanged. At a repair,

$$Z_{t+1} - Z_t = 3^{-t-1} \left((3c_t + 1) 2^{A_t} - c_{t+1} 2^{A_{t+1}} \right). \tag{1}$$

Moreover

$$\frac{Z_{t+1}}{Z_t} = \left(1 + \frac{1}{3n_t}\right) \frac{1 - c_{t+1}/n_{t+1}}{1 - c_t/n_t}. \tag{2}$$

The domain assumptions bound this ratio above and away from zero uniformly. With at most R repairs, $Z_t \asymp_{C,R} n_0 + 1$. If all pre- and post-repair values tend to infinity along a family, all snapshot ratios Z_t/Z_0 tend to 1.

4.1 Reductions

Prove the depth bound by induction on R. Without repairs, $D(n_h - c_h) = 3^h 2^{-B_h} D(n_0 - c_0)$. The endpoint difference is an even 2-adic integer, so $v_2(D(n_0 - c_0)) \geq B_h + 1$. Its nonzero integral numerator has absolute value $O_C(n_0 + 1)$, proving the base case.

From any fixed positive starting value there is a finite maximum length of an admissible prefix with at most R repairs. Indeed a good run starting at n with reference c has length at most $v_2(D(n - c)) - 1$: each good step decreases that valuation by $b(c) \geq 1$. There are finitely many reference choices and possible first-repair locations. Induction over the remaining repairs proves the assertion, uniformly on finite sets of starting values.

If the bound fails for fixed $\varepsilon > 0$, choose a family with

$$B_h - (1 + \varepsilon) \log_2(n_0 + 1) \longrightarrow \infty. \tag{3}$$

Replace each reference path by one needing the fewest repairs. Two paths with at most R repairs have the same reference depths outside the union of their repair sets. Their B_h differ by at most $2R \max b$, so failure persists. The fixed-start assertion permits a subsequence with $n_0 \to \infty$. Every post-repair value tends to infinity as well: otherwise a post-repair value stays in a finite set; the prefix before that repair has at most $R - 1$ repairs and obeys induction, while its bounded-start suffix has bounded length, contradicting (3). Pre-repair values tend to infinity since $n_{t+1} \leq (3n_t + 1)/2$.

Pass to a subsequence with a fixed positive number of repairs, fixed reference labels at repairs, and each successive repair gap either fixed or tending to infinity. Group repairs separated by fixed gaps into clusters $[\ell_i, k_i)$, $1 \leq i \leq s$, with fixed widths $w_i = k_i - \ell_i$. Distinct clusters separate by gaps tending to infinity. Set $k_0 = 0$ and $\delta_i = Z_{k_i} - Z_{k_{i-1}}$. There are no changes of Z between clusters.

Lemma 3. *Every cluster increment δ_i is nonzero.*

Proof. A zero increment says that the affine word of actual exponents across the cluster sends c_{ℓ_i} to c_{k_i}. Every intermediate rational has denominator dividing D times a power of 2 by forward iteration, and dividing D times a power of 3 by backward iteration. Thus its denominator divides D. Backward iteration from the odd endpoint shows all intermediates are odd 2-adic integers. Every exponent is consequently the exact valuation at its preceding intermediate. The word follows g from c_{ℓ_i}, hence stays in C. Replacing the internal references by this trajectory removes repairs and leaves both endpoints unchanged, contradicting minimality. □

4.2 An inequality at three places

Let $k = k_s$, $d = s + 1$, and take the positive integer vector

$$X_i = D3^k Z_{k_i} \quad (0 \leq i \leq s), \qquad H = \max_i X_i \ll 3^h (n_0 + 1).$$

At infinity use the independent forms $X_1 - X_0, \ldots, X_s - X_{s-1}, X_s$; at 2 and 3 use the coordinate forms. Their product satisfies

$$\Pi(X) \ll (n_0 + 1) 2^{-B_h}. \tag{4}$$

Here constants may depend on the fixed cluster widths. To verify it, put $v_i = v_2(D(n_{k_i} - c_{k_i})) \geq 1$. Then

$$|X_i|_2 = 2^{-A_{k_i} - v_i}, \qquad |X_i|_3 \leq 3^{-(k - k_i)}.$$

Summing (1) across a cluster and using $a_t \geq 1$ gives

$$|\delta_i| \ll 2^{A_{k_i}} 3^{-k_i} (3/2)^{w_i}.$$

Indeed a summand is at most a constant times $2^{A_{t+1}}3^{-t-1}$; its ratio to $2^{A_{k_i}}3^{-k_i}$ is at most $(3/2)^{k_i-t-1}$. Each snapshot valuation pays for the depth of the following good gap, leaving at least one unit at its endpoint. These gaps cover the complement of the clusters, so

$$\sum_{i=0}^{s} v_i \geq B_h - (\max b) \sum_i w_i + d.$$

Multiplying the real and 2-adic and 3-adic estimates cancels every A_{k_i} and k_i factor. The final real form contributes $Z_{k_s} \ll n_0 + 1$. This proves (4).

Since $h \leq B_h$, failure (3) implies $\Pi(X) \leq H^{-\eta}$ eventually for some fixed $\eta > 0$. More explicitly, writing $L = \log_2(n_0 + 1)$ and $\alpha = \log_2 3$ gives $\log_2 H \leq \alpha B_h + L + O(1)$ and $\log_2 \Pi \leq L - B_h + O(1)$; the ratio $(B_h - L)/(\alpha B_h + L)$ stays bounded away from zero. The Subspace Theorem applies to these fixed forms in dimension d.

4.3 Eliminating the exceptional subspaces

One proper rational subspace contains infinitely many vectors. Thus fixed rational coefficients z_i, not all zero, satisfy $\sum_{i=0}^{s} z_i Z_{k_i} = 0$ on an infinite subfamily. By (2) and the reductions, $Z_{k_i}/Z_0 \to 1$; hence $\sum z_i = 0$. Rewriting in increments,

$$\sum_{i=1}^{s} q_i \delta_i = 0, \qquad q_i = \sum_{j=i}^{s} z_j, \tag{5}$$

with some $q_i \neq 0$. Expand (5) by (1), omitting clusters with zero q_i. This is a zero sum of at most $2R$ nonzero terms, each a fixed rational coefficient times $2^u 3^{-t-1}$. Partition them into minimal vanishing subsums; there are finitely many partitions, so fix one on a subsequence. By S-unit finiteness, fix all within-subsum projective ratios on a further subsequence. For terms at times t, t', the 3-adic valuation of their ratio is $-t + t'$ plus a fixed coefficient valuation. A fixed ratio forces $t - t'$ bounded. No minimal subsum therefore crosses between distinct clusters. Summing the subsums in a participating cluster gives $q_i \delta_i = 0$, contradicting Lemma 3. The depth bound follows.

Finally a path with at most R repairs consists of at most $R + 1$ genuine periodic pieces and R repair positions. Each piece has depth at least β times its length minus a constant depending on C. Thus $B_h \geq \beta h - O_{C,R}(1)$, proving the length bound.

5 Scope and prior literature

This is an initial-prefix bound for each fixed repair budget. Its leading coefficient is sharp already at reference -1, but no effective constant is obtained. It does not control a growing repair budget, establish positive density of deep steps, or prove Collatz convergence.

Brox [3] proves finiteness results for cycles with few descents using logarithmic forms. Those cycle results are not the finite-prefix assertion here. Sharpe's research repository [4] studies necessary growth of parity shift defects; that count should not be identified with this repair count without a separate translation. Subspace methods themselves have appeared in Collatz research, for example Fleckinger and Abdoulkarim [5]. No new Diophantine theorem or first use of these methods for Collatz is claimed.

Relative to these named results, the additional statement is the explicit finite-prefix estimate for fixed deep-step count, its fixed finite periodic-dictionary extension, and removal of exceptional subspaces. A comprehensive priority claim needs a broader specialist literature review.

References

[1] Y. Bilu, The many faces of the Subspace Theorem (after Adamczewski, Bugeaud, Corvaja, Zannier…), *Astérisque* 317 (2008), 1–38. arXiv:0907.2098.

[2] J.-H. Evertse, H. P. Schlickewei and W. M. Schmidt, Linear equations in variables which lie in a multiplicative group, *Ann. of Math.* 155 (2002), 807–836. arXiv:math/0409604.

[3] T. Brox, Collatz cycles with few descents, *Acta Arith.* 92 (2000), 181–188. doi:10.4064/aa-92-2-181-188.

[4] M. Sharpe, Collatz research repository, especially `paper/defects.tex`, accessed 21 October 2026. `https://github.com/msharpe248/collatz`.

[5] J.-J. Fleckinger and A. Abdoulkarim, Sur la conjecture de Collatz, preprint (2016). arXiv:1607.02370.